SWAYAM SAMPAD DAS

Data Engineer | SQL · Python · ETL/ELT · Data Warehousing & Modeling · BigQuery · PySpark · Cloud Data Platforms (Azure, GCP, AWS)

Bangalore | +91 93111 47087 | s3091990@gmail.com | linkedin.com/in/swayamsampad-das-b0aa9090 | github.com/ssdas1309 | Immediate joiner

SUMMARY

Data Engineer with **4.5 years** of experience designing **ETL/ELT pipelines, data warehouses and data models** with **SQL and Python** (CTEs, window functions, query tuning, partitioning) on BigQuery, ClickHouse, Azure Databricks and an Apache Iceberg lakehouse. Delivered enterprise-scale data solutions in FinTech (AEPS, UPI, BBPS): led the lakehouse migration that cut processing time by **30%**, orchestrated **1000+ daily reports** with Airflow, and shipped a reconciliation platform whose pipelines run **4x faster**. Learning Snowflake; warehouse experience so far is on BigQuery.

EXPERIENCE

iServeU — Fintech company running payment infrastructure (AEPS, UPI, MATM, BBPS) for banks and agent networks across India

Software Engineer 2 (Python Team Lead) *May 2024 – Oct 2026*

- Ran the company's batch data platform on **Azure Databricks**, building **PySpark** jobs for multi-product payment reconciliation; later led its migration to an on-prem **Apache Iceberg** lakehouse (Spark + Trino + Nessie), cutting data processing time by **30%** and improving system performance by **25%**.
- Designed **Azure Data Factory (ADF)** ingestion and orchestration as part of the planned BigQuery → Azure Databricks migration, mapping existing Airflow/BigQuery pipelines (**1000+ daily reports**) to ADF pipelines and Databricks jobs.
- Built **Node.js** services that consume transaction events from **GCP Pub/Sub** and insert them into **BigQuery**; later re-engineered the service in **Go** with automatic schema alignment and dead-letter handling for high-throughput real-time ingestion.
- Built **Dataflow** streaming pipelines that read transaction events from **GCP Pub/Sub** and write them into **Apache Iceberg tables on BigLake**, replacing the earlier insert path with managed, auto-scaling ingestion.
- Ran **Dataproc** (Spark) jobs to read, transform and maintain **Apache Iceberg tables on BigLake**, so the same lakehouse data could be queried from BigQuery and Spark.
- Used **Google Cloud Storage (GCS)** as the landing, cache and delivery layer: SFTP switch files staged to GCS (upload time down **50%**), Parquet caches for Airflow checkpoint/resume, delivery of **1000+ daily reports**, and GCS Parquet loaded into the Iceberg lakehouse with Spark.
- Extended the **Pub/Sub** pipeline with dead-letter handling and automatic schema alignment, so malformed transaction events no longer block real-time ingestion into **BigQuery**.
- Built **100+ Looker dashboards** for the finance team on top of BigQuery and reconciliation data, giving finance self-serve views of settlements, mismatches and daily reports.
- Architected **ReconHub**, a config-driven reconciliation platform for AEPS, UPI and MATM — **React/Next.js 14** frontend, **FastAPI** backend and **Polars** pipelines running **4x faster** than the earlier Pandas version.
- Built a **ClickHouse** reporting layer (ReplacingMergeTree, partitioned by recon date and cycle) with a FastAPI query API serving real-time reconciliation and mismatch analytics.
- Designed a **Trino**-based query management API, cutting query execution time by **45%** and raising internal user adoption by **60%**.
- Automated commission reconciliation with cycle-aware **Apache Airflow** DAGs — IST/UTC resolution, checkpoint/resume via PostgreSQL and Redis, GCS Parquet caching for fault tolerance.
- Built **GCP Vertex AI** solutions for KYC document OCR and transaction fraud detection; migrated OCR from GCP Vision API to AWS Textract, improving accuracy by **20%** and cutting API costs by **35%**.
- Built **Jira and GitLab MCP servers** and a **Qdrant RAG** assistant over engineering docs (LangChain), reducing development time by **20%**.
- Led a team of Python engineers — design reviews, code reviews and mentoring; raised team productivity by **20%** and cut deployment errors by **25%**.

Software Engineer *May 2023 – May 2024*

- Automated multi-product reconciliation (AEPS, UPI, MATM) with **Apache Spark (PySpark)**, Pandas and NumPy, reducing reconciliation time by **60%** and improving data accuracy by **25%**.
- Designed an **Airflow DAG factory** orchestrating **1000+ daily BigQuery reports** — SQL-file driven, dynamic DAG registration, cycle-aware scheduling and automated GCS delivery.
- Built REST APIs for transaction clearing automation with Flask and FastAPI, cutting processing time by **45%**.
- Automated SFTP → GCS ingestion of switch files, reducing upload time by **50%** and data errors by **20%**; implemented KYC OCR that sped up document processing by **70%**.

Associate Software Engineer / Trainee *Feb 2022 – May 2023*

- Built **Jenkins CI/CD** pipelines for Cloud Run and VM deployments; containerised Go, Java, **Node.js** and Tomcat services with **Docker**.
- Wrote **Terraform** IaC and Ansible playbooks for repeatable cloud provisioning; ran log pipelines into the ELK Stack.

SKILLS

Data Engineering: PySpark / Apache Spark (Azure Databricks), Apache Airflow, Apache Iceberg, Trino, Azure Data Factory (ADF), Polars, Pandas, ETL/ELT, batch pipelines, SFTP ingestion, data quality & validation, Spark/SQL performance tuning

Full Stack: React, Next.js 14, Node.js, TypeScript, FastAPI, Flask, REST, tRPC

Cloud: Microsoft Azure (Databricks, ADF), Google Cloud (BigQuery, BigLake, Dataproc, Dataflow, Pub/Sub, GCS, Looker, Cloud Run, Vertex AI), AWS (Textract)

Databases: PostgreSQL, BigQuery, ClickHouse, Redis, Cassandra; Snowflake (learning, not production yet)

Languages: Python, SQL, Go, TypeScript/JavaScript, Bash

DevOps & AI: Jenkins CI/CD, Docker, Terraform, Git/GitLab, Kubernetes (basics); LangChain, LangGraph, RAG, Qdrant, MCP servers

EDUCATION & CERTIFICATIONS

B.Tech, Electrical and Electronics Engineering — Veer Surendra Sai University of Technology (VSSUT), Burla, 2018–2022

Certifications: Coursera / DeepLearning.AI — ML & AI Foundations; Advanced Predictive Modeling; Intro to TensorFlow; Structuring ML Projects